

AN EMPLOYEE REQUESTS INSTALLATION OF LICENSED SOFTWARE THROUGH THE SERVICE CATALOG

Complete Project Documentation

ServiceNow – Service Portal, Service Catalog, Flow Designer, Approvals, Catalog Tasks, Notifications and Update Sets

Project	An Employee Requests Installation of Licensed Software through the Service Catalog
Platform	ServiceNow
Primary Interface	Service Portal
Workflow	Catalog Request → REQ/RITM → Approval → Catalog Task → Fulfillment → Status Tracking → Notification

1. Project Overview

The project focuses on developing an automated Licensed Software Installation Request Management system using ServiceNow. The solution provides employees with a simple and structured way to request software installation through the Service Catalog.

The main purpose is to capture software requirements accurately, obtain the required approval, create a fulfillment task for the software support team, and provide request status tracking throughout the process.

The system uses Service Catalog, Catalog Item, Catalog Variables, Flow Designer, Approval Requests, Catalog Tasks, and Update Sets to manage and transfer the software request configuration.

The screenshot shows the ServiceNow interface for configuring a Catalog Item. The browser address bar indicates the URL: `dev270509.service-now.com/now/nav/ui/classic/params/target/sc_cat_item.do%3Fsys_id%3D4130260847530310d799bb30116d4337%26sysparm_record_target%3Dsc_cat_item%26sysparm_record_row...`. The page title is "Catalog Item - Software Installation Request".

Below the title bar, there is a blue informational box stating: "Catalog items are goods or services available to order from the service catalog. Items can be anything from hardware, like tablets and phones, to software applications, to furniture and office supplies." followed by two bullet points: "Enter a Name and Short description to display for the item." and "Enter a Price, approvals, variables, and other information as needed."

The configuration form includes the following fields:

- Name:** Software Installation Request
- Application:** Global
- Active:** ☒
- Fulfillment automation level:** Unspecified
- Category:** Software
- State:** Published
- Checked out:** false
- Owner:** System Administrator
- Secondary owners:** (empty)

Below the form, there are tabs for "Item Details", "Process Engine", "Picture", "Pricing", and "Portal Settings". The "Item Details" tab is active, showing a "Short description" field with the text "Request to install company-approved software" and a "Description" field with a rich text editor toolbar.

Figure 1: Software Installation Request – Catalog Item Configuration

Project Workflow

The implemented process follows this sequence:

1. Employee opens the **Software Installation Request** catalog item through the Service Portal.

2. Employee enters **Software Name, Version Required, License Justification, and Urgency**.
3. Employee submits the request.
4. ServiceNow creates the **Request (REQ)** and **Requested Item (RITM)** records.
5. Flow Designer triggers the **Software Request Workflow**.
6. The request is sent for approval.
7. Once approved, the flow creates a **Catalog Task (SCTASK)**.
8. The task is assigned to the available **Software** support group.
9. The task can be processed and its state updated.
10. The request can be tracked through ServiceNow.

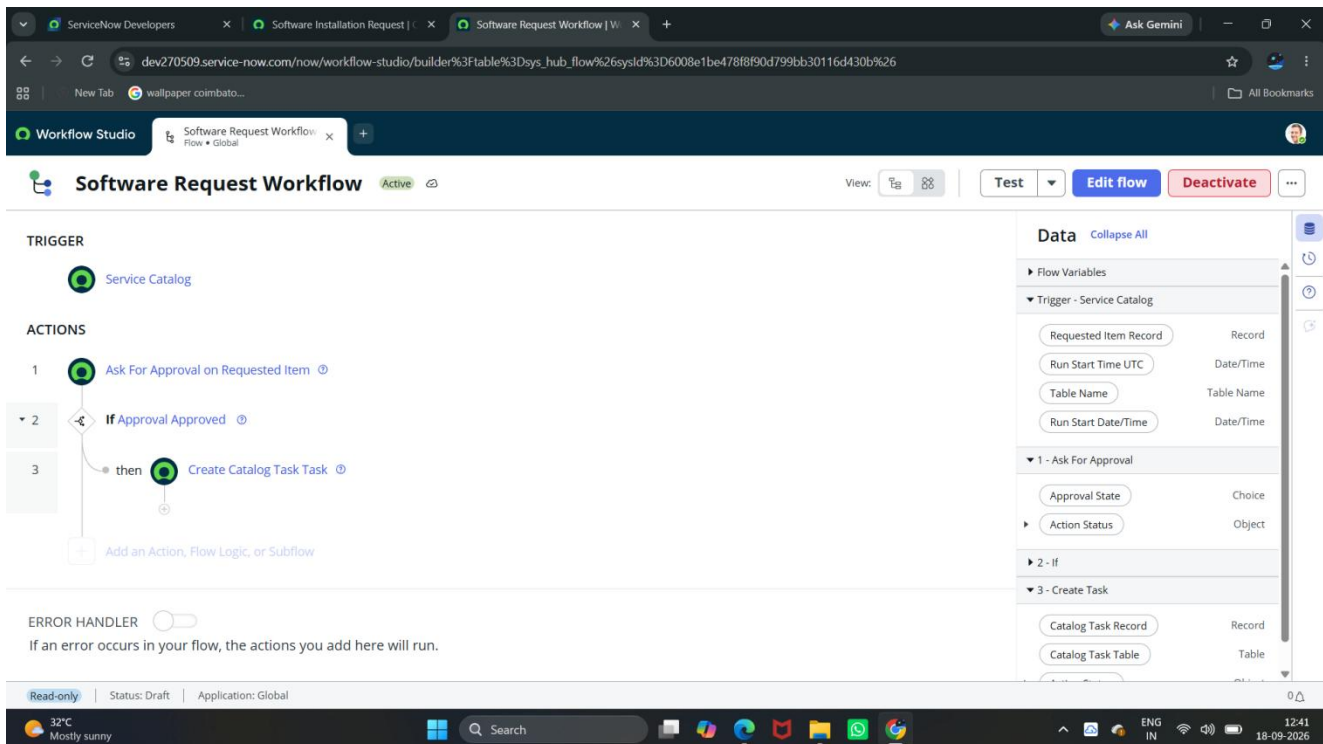


Figure 2: Software Request Workflow – Approval and Catalog Task Automation

Project Phases

The project was implemented through the following phases:

1. Requirement Analysis & Planning – Identifying the need for employees to request software through the Service Catalog and defining the required software, version, business justification, urgency, approval, and fulfillment requirements.

2. Service Catalog & Form Configuration-Creating the Software Installation Request catalog item and configuring the catalog variables required to capture software request information.

3. Workflow Automation & Approval-Using Flow Designer to automate approval processing and create a Catalog Task after approval.

4. Testing & Validation-Testing the catalog request, REQ/RITM generation, approval flow, Catalog Task creation, task state, and flow execution.

5. Deployment, Documentation & Final Presentation-Completing the Update Set, exporting and importing the configuration, resolving preview issues, committing the Update Set, validating the target instance, and documenting the completed project.

Final Outcome

The completed project provides a structured and automated process for employees to request software installation through the ServiceNow Service Catalog.

Employees submit a software request by providing the required software details, version information, business justification, and urgency. The request is processed through Flow Designer for approval, and once approved, a Catalog Task is automatically created for the Software support team.

The solution provides structured request tracking through REQ, RITM, and SCTASK records and uses Update Sets to transfer the completed configuration between ServiceNow instances.

The screenshot displays the ServiceNow Catalog Task form for SCTASK0010009. The form is divided into several sections:

- Header:** Includes the task number (SCTASK0010009) and a status bar with buttons for 'Follow', 'Update', 'Close Task', and 'Delete'.
- Form Fields:**
 - Number:** SCTASK0010009
 - Assigned to:** Searchable field.
 - Configuration item:** Searchable field.
 - Active:** Checked checkbox.
 - Approval:** Not Yet Requested (dropdown).
 - Priority:** -- None -- (dropdown).
 - State:** Open (dropdown).
 - Request item:** Searchable field.
 - Requested for:** Searchable field.
- Description:** Short description: Software Fulfillment Task. Description: (empty text area).
- Work notes:** Work notes (empty text area). Includes a 'Comments (Customer visible)' checkbox and a 'Post' button.
- Activities:** 1 activity listed: System Administrator. Details: Impact: 3 - Low, Opened by: System Administrator, State: Open. Field changes: 2026-09-30 11:24:21.
- Assignment group:** Software (searchable field).

At the bottom, there are buttons for 'Update', 'Close Task', and 'Delete'.

2. Phase 1 – Requirement Analysis & Planning

Step 1: Understand the Project Requirement

The main requirement is to provide employees with a centralized ServiceNow Service Catalog item through which they can request installation of licensed software.

The system should:

- Allow employees to submit a licensed software installation request.
- Capture **Software Name, Version Required, License Justification, and Urgency**.
- Route the request for approval.
- Automatically create a Catalog Task after approval.
- Allow the employee to track the request status.
- Maintain the request information through REQ, RITM, and Catalog Task records

Step 2: Identify Stakeholders

The major stakeholders involved in the process are:

- **Employee / Requester** – submits the software installation request.
- **Approver** – reviews and approves the request.
- **Software Support Team** – processes the generated Catalog Task.
- **ServiceNow Administrator** – configures the catalog item, flow, approvals, and deployment.

Step 3: Define ServiceNow Components

The following ServiceNow components are planned for the project:

- Service Catalog
- Software Installation Request Catalog Item
- Catalog Variables
- Flow Designer
- Approval
- Catalog Task (`sc_task`)
- Request (`REQ`)
- Requested Item (`RITM`)
- Update Sets

Step 4: Define the Request Process

The planned request process is:

Employee → Service Portal → Software Installation Request → Enter Details → Submit → REQ/RITM
→ Approval → Catalog Task → Fulfillment → Status Tracking

Step 5: Define Functional Requirements

The system should provide the following functionality:

- Employee can access the Service Catalog.
- Employee can open the Software Installation Request item.
- Employee can enter software name and required version.
- Employee can provide license justification and urgency.
- The request generates a REQ and RITM after submission.
- The request is routed for approval.
- A Catalog Task is created after approval.
- The Software support team can process the task and update its state.
- The requester can track the request status.

- The complete request remains traceable through REQ, RITM, and SCTASK records

Step 6: Project Planning

The overall project is planned in the following sequence:

Requirement Analysis & Planning → Service Catalog & Form Configuration → Workflow Automation & Approval → Testing & Validation → Deployment, Documentation & Final Presentation

3. Phase 2 – Service Catalog & Form Configuration

The second phase of the project focuses on creating and configuring the **Software Installation Request** in the ServiceNow Service Catalog. This phase converts the requirements identified in Phase 1 into an actual catalog item and request form.

2.1 Create the Service Catalog Item

A dedicated Service Catalog item was created for the software installation request.

Catalog Item: Software Installation Request

The catalog item acts as the entry point for employees to request software installation. It captures the software details, business justification, and urgency required to process the request.

The catalog item is connected to the subsequent request and workflow process so that the submitted information can be used for validation, approval, license checking, task creation, and fulfillment.

ServiceNow Developers | Software Installation Request | Ask Gemini

dev270509.service-now.com/now/nav/ui/classic/params/target/sc_cat_item.do%3Fsys_id%3D4130260847530310d799bb30116d4337%26sysparm_record_target%3Dsc_cat_item%26sysparm_record_row...

servicenow | All | Favorites | History | Workspaces | Catalog Item - Software Installation Request

Catalog items are goods or services available to order from the service catalog. Items can be anything from hardware, like tablets and phones, to software applications, to furniture and office supplies.

- Enter a Name and Short description to display for the item.
- Enter a Price, approvals, variables, and other information as needed.

Name: Software Installation Request

Application: Global

Active: ☒

Fulfillment automation level: Unspecified

Category: Software

State: Published

Checked out: false

Owner: System Administrator

Secondary owners:

Item Details | Process Engine | Picture | Pricing | Portal Settings

Short description: Request to install company-approved software

Description:

32°C Mostly sunny | 12:37 18-09-2026

Figure 1: Software Installation Request – Catalog Item Configuration

2.2 Configure Catalog Variables

The request form was configured with variables to capture the information required by the Software Support Team.

The following variables were configured based on the implemented form:

Variable	Purpose
Software Name	Captures the software required for installation
Version Required	Captures the required software version
License Justification	Captures the reason for requesting the software
Urgency	Captures the urgency level of the request

The variables ensure that the request contains sufficient information before it enters the automated approval and fulfillment process.

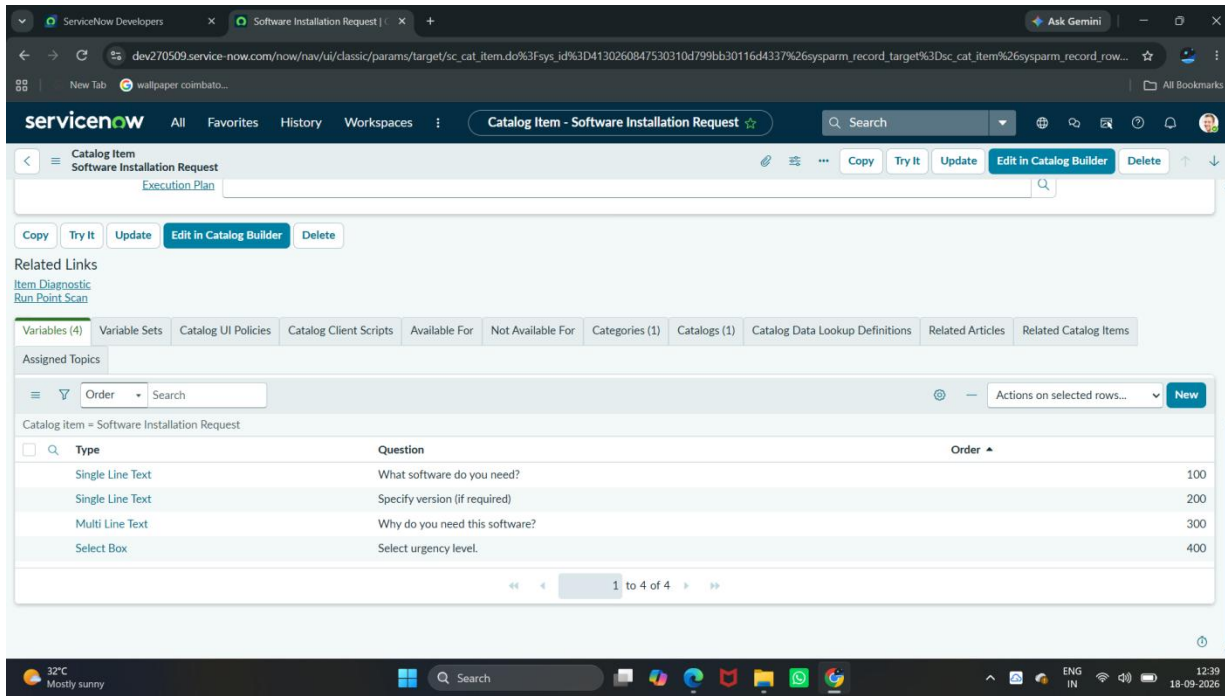


Figure 2: Software Installation Request – Configured Catalog Variables

2.3 Configure the Request Form

The Service Catalog form was designed to make the software request process simple and structured for employees.

The form captures the following major categories of information:

Software Information

- Software Name
- Version Required

Business Information

- License Justification

Priority Information

- Urgency

This information is later consumed by the workflow to determine the appropriate approval and fulfillment path.

The image shows a web browser window displaying the ServiceNow 'Software Installation Request' form. The browser's address bar shows a URL starting with 'https://dev438159.service-now.com'. The ServiceNow logo is in the top left, and navigation links like 'Knowledge', 'Catalog', 'Requests', 'Approvals', 'System Status', 'Cart', 'Tours', and 'System Administrator' are in the top right. The breadcrumb trail is 'Home > Service Catalog > Software > Software Installation Request'. The form itself has a title 'Software Installation Request' and a subtitle 'Request to install company-approved software'. Below this is a description: 'Allows employees to request software installations. Fulfills via Software Support Team.' The form contains several input fields: 'What software do you need?' (a large text area), 'Specify version (if required)' (a text area), 'Why do you need this software?' (a large text area), and 'Select urgency level.' (a dropdown menu currently set to 'Critical'). There is also an 'Add attachments' section. On the right side of the form, there is a sidebar with 'Quantity: 1', 'Delivery Time: 0 Days', and buttons for 'Add to Cart', 'Save as Draft', and 'Order Now'. A chat icon is visible in the bottom right corner of the form area.

Figure 3: Software Installation Request - Service Portal Form

2.4 Form Validation and Mandatory Fields

Required fields were configured to ensure that employees cannot submit an incomplete software request.

The request form captures the key software information required for processing, including Software Name, Version Required, License Justification, and Urgency. These fields provide the information required for the approval and fulfillment process.

This validation improves data quality and prevents incomplete requests from entering the approval and fulfillment process.

2.5 Configure User-Friendly Form Behavior

The Service Catalog form provides a simple and structured request experience. The form allows the requester to enter software details, provide a business justification, and select the urgency level before submitting the request.

The Service Catalog form was configured to provide a structured and user-friendly experience.

The form configuration supports:

- Selection of urgency.

- Entry of business justification.
- Selection of the employee/requested-for information.
- Identification of the installation device.
- Required-date selection.

The collected information is carried forward to the **Requested Item (RITM)** and subsequently becomes available to the automated workflow.

2.6 Request and Requested Item Generation

When the employee submits the catalog form, ServiceNow generates the corresponding Request (REQ) and Requested Item (RITM) records.

The process follows:

Employee → Service Catalog Item → Catalog Variables → Submit → Request (REQ) → Requested Item (RITM)

The RITM stores the submitted catalog request information and becomes the Requested Item record used by the approval and fulfillment flow.

The workflow then uses this information for:

- Approval processing
- Catalog task creation
- Fulfillment
- Status updates
- Notifications

Requested Item
RITM0010002 View: Self Service

Number: RITM0010002
Due date: 2026-09-04 21:47:16

Approval: Approved
Quantity: 1
Stage: Assess or Scope Task

Delivery Plan Status for: Request installation of approved and licensed software.

Task	Duration	Expected Start	Started	Completed
Assess or Scope Task	1 Day	2026-09-03	2026-09-03	
Provide requested service	1 Day	2026-09-04		

Description

Employees can use this service to request installation of approved and licensed software on their organizational device. Requests are subject to the required approval and fulfillment process.

Figure 4: Generated Requested Item with Software Installation Request Details

2.7 Integration with the Workflow

The configured catalog item is connected to the **Software Request Workflow** through the Catalog Item process engine.

The implemented process is:

Catalog Item Requested → Ask For Approval → If Approval Approved → Create Catalog Task

After approval, the flow creates a Catalog Task for software fulfillment.

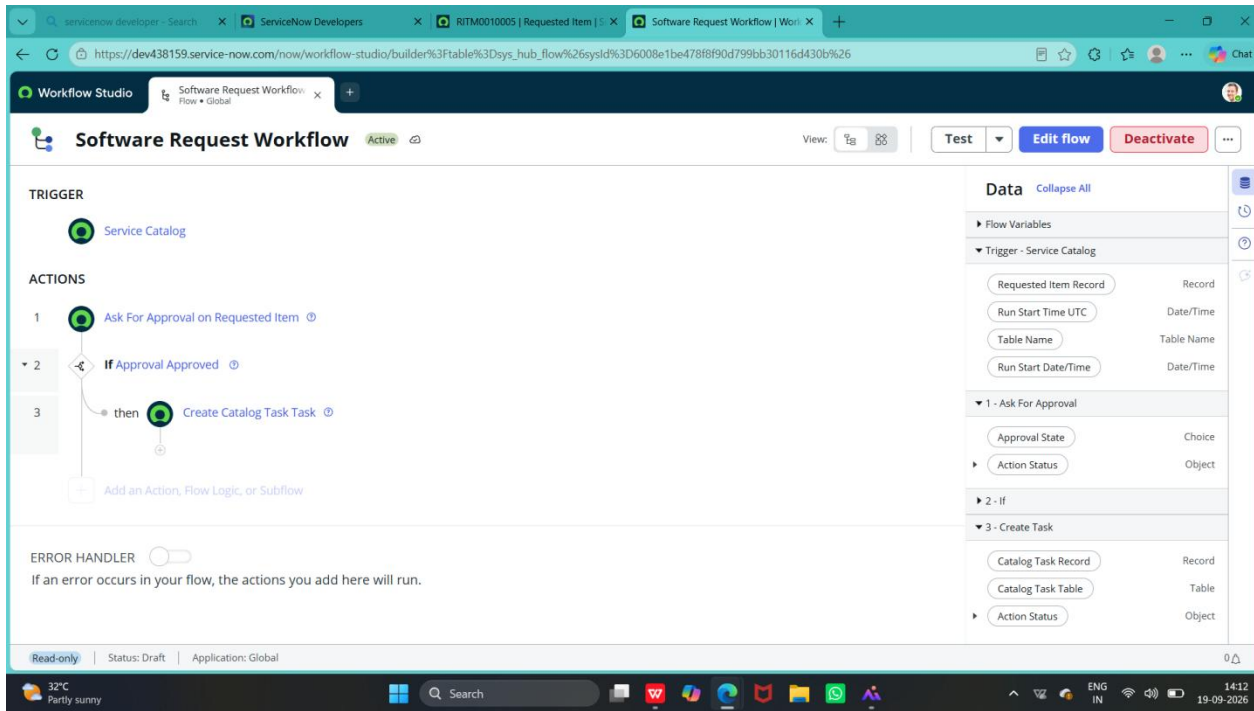


Figure 5: Software Request Workflow - Approval and Catalog Task Automation

2.8 Example of Configured Request

An example request was successfully created with values similar to:

- **Software Name:** Microsoft Office
- **Version Required:** Microsoft Office 2024
- **License Justification:** Required for academic and office-related work
- **Urgency:** Normal

This confirms that the Service Catalog form is successfully capturing the information required for downstream processing.

The screenshot shows a ServiceNow web interface for a 'Software Installation Request'. The breadcrumb trail is 'Home > Service Catalog > Software > Software Installation Request'. The form title is 'Software Installation Request' with a subtitle 'Request to install company-approved software'. A description states: 'Allows employees to request software installations. Fulfills via Software Support Team.' The form contains four main input sections: 'What software do you need?' with the value 'Microsoft Office 365', 'Specify version (if required)' with the value '2021', 'Why do you need this software?' with the value 'Required for daily work', and 'Select urgency level.' with the value 'High'. To the right of the form is a sidebar with 'Quantity: 1', 'Delivery Time: 2 Days', and three buttons: 'Add to Cart', 'Save as Draft', and 'Order Now'.

Figure 6: Example Software Installation Request with Submitted Variables

Phase 3 – Workflow Automation & Approval

The third phase focuses on improving the **user interface and user experience** of the Licensed Software Installation Request in ServiceNow. The objective is to make the catalog request form simple, clear, and easy for employees to complete while ensuring that the required information is captured correctly.

3.1 Form Layout and Field Organization

The Licensed Software Installation Request form is organized around the information required to process a software installation request.

Software Information

- **Software Name**
- **Version Required**

Business Information

- **License Justification**

Priority Information

- **Urgency**

This arrangement helps the requester provide information in a logical sequence.

3.2 Mandatory Field Validation

The catalog item captures the key information required for processing the software request:

Software Name,
Version Required,
License Justification,
Urgency.

This prevents incomplete requests from entering the workflow.

3.3 User-Friendly Field Configuration

Appropriate variable types are used according to the information being captured.

For example

Field	UI Purpose
Software Name	Enter the software required
Version Required	Enter the required software version
License Justification	Enter the business reason for the request
Urgency	Select the urgency level

3.4 Example of Customized Request Form

A completed request demonstrates how the customized UI captures the information required for processing.

Example values include:

- **Software Name:** Microsoft Office
- **Version Required:** Microsoft Office 2024
- **License Justification:** Required for academic and office-related work
- **Urgency:** Normal

The screenshot shows a ServiceNow web interface for a 'Software Installation Request'. The breadcrumb trail is 'Home > Service Catalog > Software > Software Installation Request'. The form title is 'Software Installation Request' with a subtitle 'Request to install company-approved software'. A description states: 'Allows employees to request software installations. Fulfills via Software Support Team.' The form contains several input fields: 'What software do you need?' with the value 'Microsoft Office 365', 'Specify version (if required)' with the value '2021', 'Why do you need this software?' with the value 'Required for daily work', and 'Select urgency level' with a dropdown menu set to 'High'. On the right side, there is a sidebar with a 'Quantity' dropdown set to '1', a 'Delivery Time' of '2 Days', and three buttons: 'Add to Cart', 'Save as Draft', and 'Order Now'.

Figure 3.1: Software Installation Request - Configured Catalog Form

3.5 Workflow Integration

The customized form is designed to work with the subsequent automation process:

Catalog Item → Request (REQ) / Requested Item (RITM) → Approval → Catalog Task → Fulfillment

Therefore, focused on appearance but also ensures that the information collected from the employee can be consumed by the workflow.

- The Software Request Workflow is configured in Flow Designer.
- Approval processing is automated.
- Approved requests automatically create a Catalog Task.
- The task is assigned to the available Software support group.
- The request flow is traceable through REQ, RITM, and SCTASK records

3.6 Workflow Automation Overview

The workflow is designed to reduce manual processing and ensure that every software request follows a controlled process.

Employee → Catalog Request → REQ/RITM → Approval → Catalog Task → Fulfillment → Status Tracking

The workflow is implemented using the ServiceNow workflow/Flow Designer environment.

3.7 Workflow Configuration

The configured workflow is:

Software Request Workflow

The configured flow is **Software Request Workflow**. It is triggered when a Catalog Item is requested and processes the Requested Item through approval and fulfillment task creation.

The flow contains:

- Catalog Item Requested
- Ask For Approval on Requested Item
- If Approval Approved
- Create Catalog Task

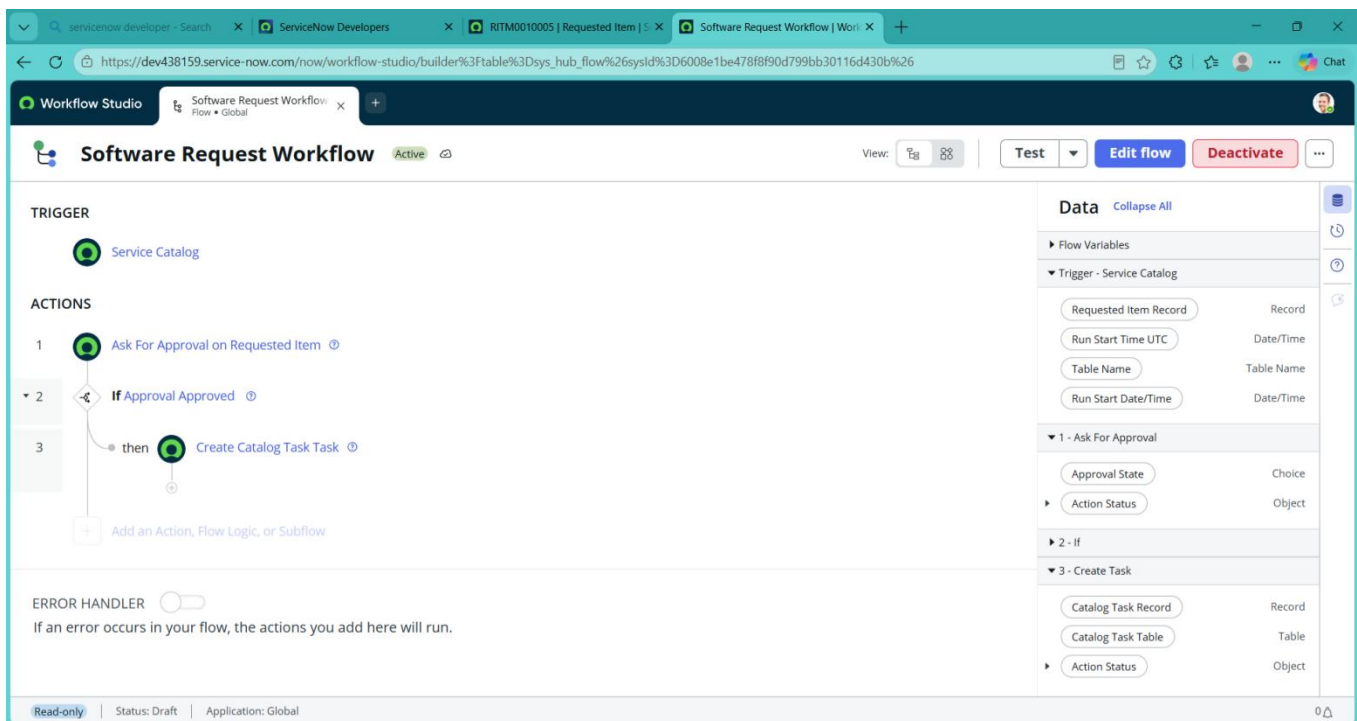


Figure 4: Software Request Workflow - Approval and Catalog Task Creation

3.8 Approval Process

After the request is submitted, the flow sends the Requested Item for approval using the **Ask For Approval** action. The configured approver reviews the request and provides the approval decision.

- Requested software
- Business justification
- Employee
- Department
- Device
- Required date
- Urgency
- License information

The approval decision determines whether the request can proceed.

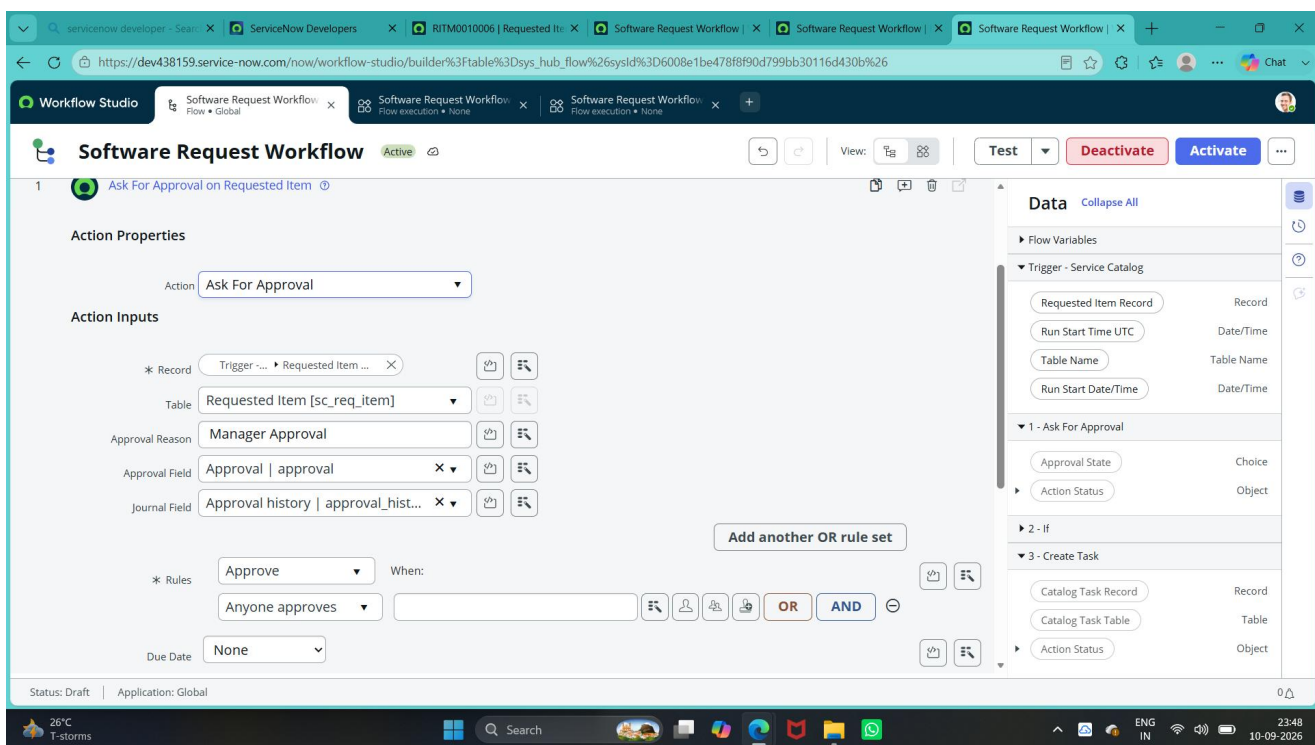


Figure 5: Software Request Workflow - Ask For Approval Configuration

3.9 Approval Decision

The flow evaluates whether the approval result is **Approved**.

If Approved: the flow continues to **Create Catalog Task**.

The request does not proceed to the task-creation step until the approval condition is satisfied.

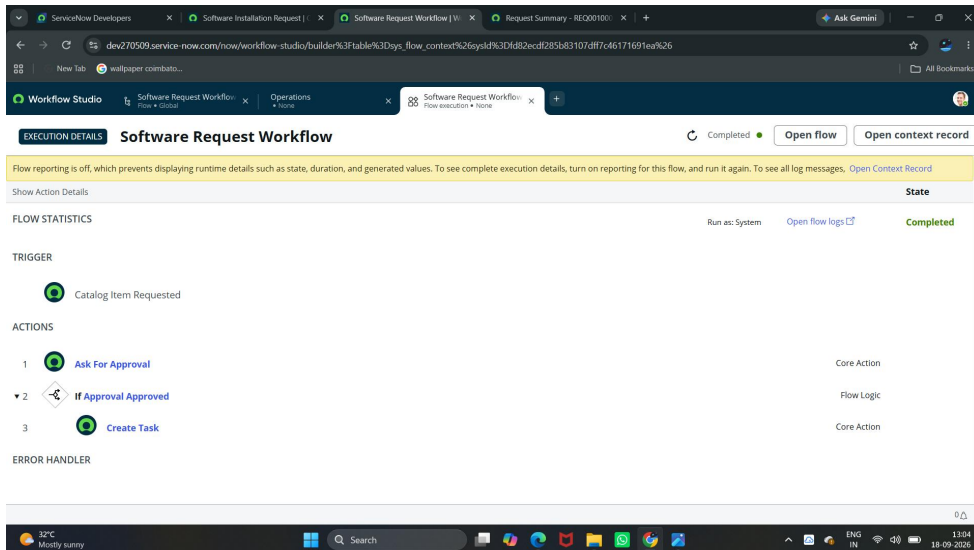


Figure 6: Software Request Workflow - Execution Detail

3.10 Catalog Task Creation

Once the request passes the required approval and license checks, the workflow creates a **Catalog Task (sc_task)** for the software support team.

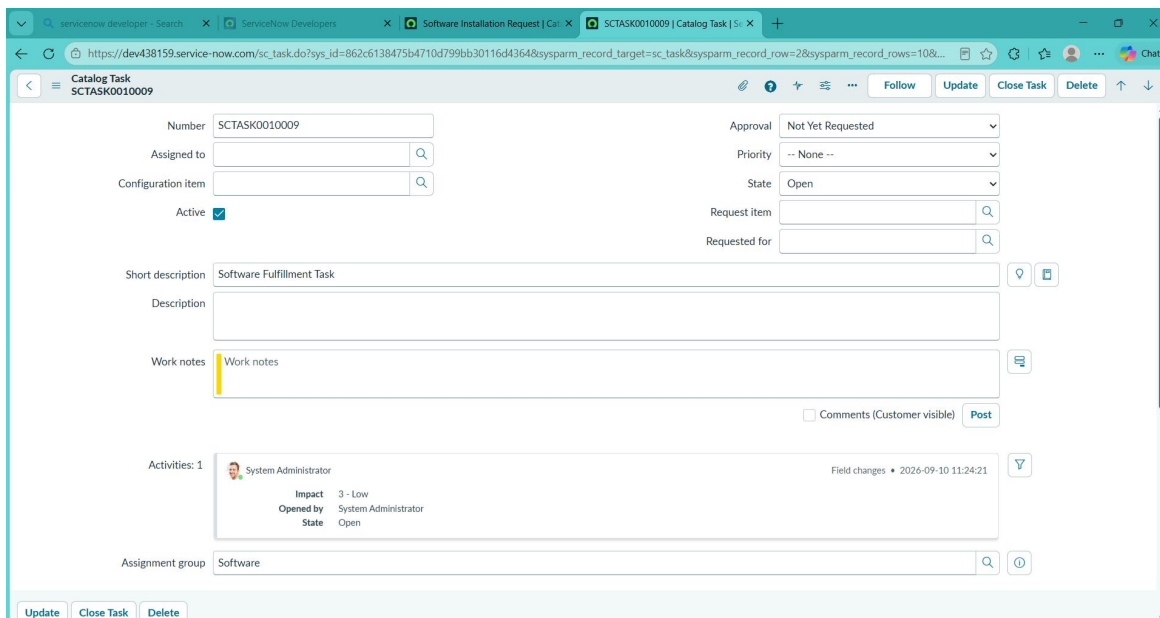
The task contains the information required to perform the installation.

The task is assigned to:

Software support group

The support team can then process the installation and update the task status.

Once the request is approved, the flow automatically creates a **Catalog Task (sc_task)** with the short description **Software Fulfillment Task**. The task is assigned to the available **Software support group** for fulfillment.



3.11 Fulfillment Process

The fulfillment process follows:

Approved Request → Catalog Task → Software Support Group → Task Processing → Task Status Update

The support team installs the requested software on the specified device and updates the catalog task.

Once fulfillment is completed, the request can move toward closure.

3.12 Notifications

ServiceNow email records generated during request processing were reviewed as part of testing to verify request and approval-related communication.

3.13 Update Set

The workflow automation configuration is captured in the update set:

Licensed Software Installation - Capstone

This allows the workflow-related configuration to be tracked and moved between ServiceNow instances as required.

The project configuration was captured in the **Licensed Software Installation - Capstone** Update Set. The Update Set was completed, exported, imported into a target instance, previewed, and committed successfully.

3.14 End-to-End Workflow

The complete automated process is:

Employee

↓

Software Installation Request

↓

REQ / RITM

↓

Ask For Approval



Approval Decision



Create Catalog Task



Software Support Group



Task Processing / Status Update

3.15 Phase 3 Outcome

At the end of Phase 3:

- The Software Request Workflow is automated.
- Approval processing is configured using Flow Designer.
- Approval decisions control the next step of the flow.
- Catalog Tasks are automatically created after approval.
- The generated task uses the Software support group for fulfillment.
- Request and task records remain traceable through REQ, RITM, and SCTASK.

Phase 4 – Testing & Validation

The fourth phase focuses on validating the **Software Installation Request** to ensure that the catalog item, request generation, approval flow, and Catalog Task creation work correctly. Testing was performed by submitting a software request and reviewing the resulting flow execution and fulfillment task.

4.1 Testing Objectives

The main objectives of this phase are:

- Verify that the catalog request form works correctly.
- Verify REQ and RITM creation.
- Verify approval routing and approval decisions.
- Confirm automatic Catalog Task creation.

- Verify assignment to the **Software Support Team**.
- Verify all configured notifications.
- Confirm request status updates and closure.
- Ensure appropriate access and security controls are applied.

4.2 Functional Testing

Functional testing verifies that each major function of the software request process operates according to the requirements.

Test Area	Expected Result
Catalog item access	Employee can access Software Installation Request
Request submission	Request is submitted successfully
REQ/RITM creation	REQ and RITM are generated
Approval	Request is sent for approval
Approval decision	Approved request continues in the flow
Catalog Task	Task is created after approval
Assignment	Task is assigned to the Software support group
Flow execution	Software Request Workflow completes successfully
Task tracking	Catalog Task can be viewed and its state updated

4.3 Catalog Form Testing

The catalog form is tested with valid and invalid inputs.

Valid Input Testing

A complete request is submitted with all required values, such as:

- Software Name: **Microsoft Office**
- Version Required: **Microsoft Office 2024**

- License Justification: **Required for academic and office-related work**
- Urgency: **Normal**

The system should successfully submit the request and create the corresponding RITM.

The screenshot shows the ServiceNow 'Software Installation Request' form. The form is titled 'Software Installation Request' and includes a description: 'Request to install company-approved software'. It allows employees to request software installations, fulfilled via the Software Support Team. The form fields include: 'What software do you need?' (Microsoft Office 365), 'Specify version (if required)' (2021), 'Why do you need this software?' (Required for daily work), and 'Select urgency level.' (High). On the right, there is a sidebar with 'Quantity: 1', 'Delivery Time: 2 Days', and buttons for 'Add to Cart', 'Save as Draft', and 'Order Now'.

Invalid Input Testing

The form is also tested by:

- Leaving mandatory fields empty.
- Entering incomplete information.
- Selecting invalid or unsupported values.
- Attempting to submit without required request details.

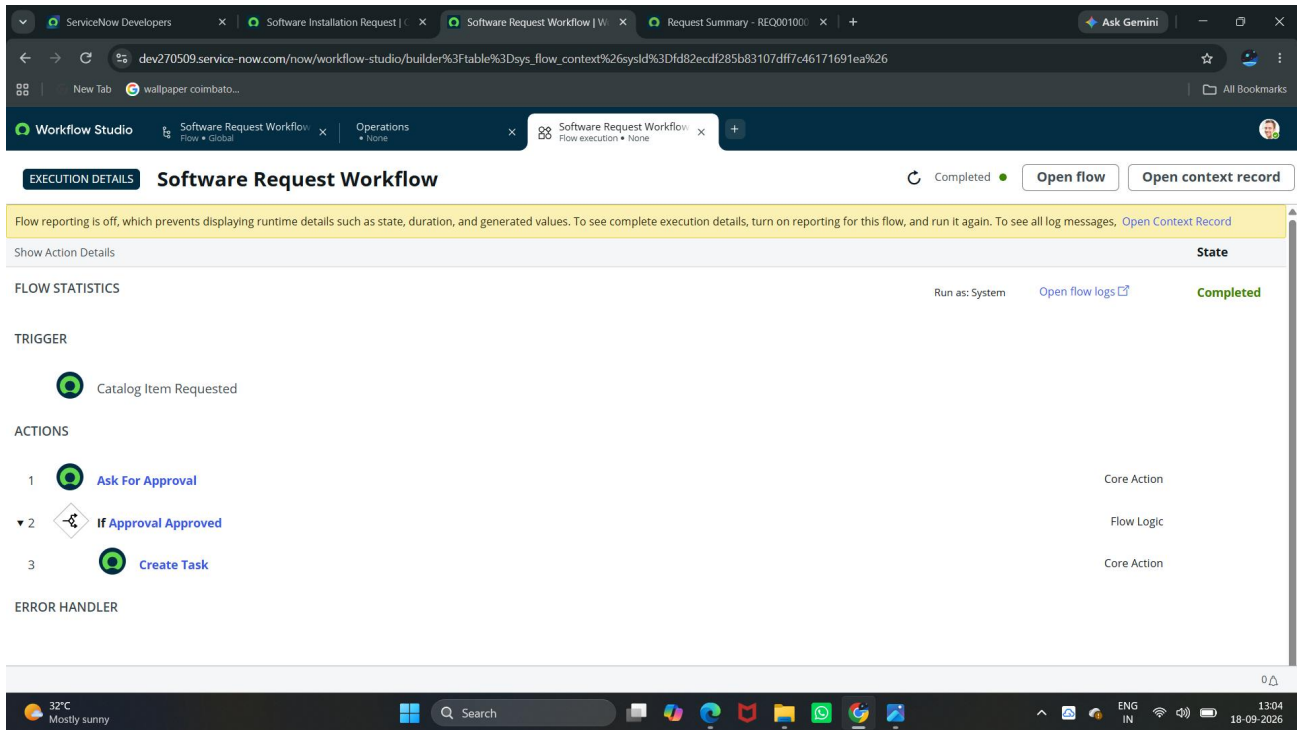
The system should prevent incomplete requests from being submitted.

4.4 Workflow Testing

The complete workflow is tested from submission through fulfillment.

Employee Submission → REQ/RITM → Approval → Catalog Task Creation

Each workflow stage is verified to ensure that the request moves to the correct next step.



4.5 Approval Testing

The approval process was tested using the configured **Ask For Approval** action. After approval, the flow evaluated the approval condition and continued to Catalog Task creation.

Approval testing verifies both possible approval outcomes.

Approved Request

Request → Approval → Approved → Catalog Task → Software Installation

The approved request should continue toward fulfillment.

Rejected Request

Request → Approval → Rejected → Requester Notification

A rejected request should not proceed to software installation.

4.6 Catalog Task Testing

After the approval was completed, the flow automatically created a Catalog Task for software fulfillment. The generated task was verified in the `sc_task` table

The notification configuration and delivery are verified to ensure that the correct recipient receives the appropriate message at the correct stage.

The screenshot shows a ServiceNow Catalog Task form for SCTASK0010009. The form is titled 'Catalog Task' and 'SCTASK0010009'. It includes several input fields and dropdown menus. The 'Number' field is filled with 'SCTASK0010009'. The 'Assigned to' field is empty. The 'Configuration item' field is empty. The 'Active' checkbox is checked. The 'Approval' dropdown is set to 'Not Yet Requested'. The 'Priority' dropdown is set to '-- None --'. The 'State' dropdown is set to 'Open'. The 'Request item' field is empty. The 'Requested for' field is empty. The 'Short description' field is filled with 'Software Fulfillment Task'. The 'Description' field is empty. The 'Work notes' field is empty. The 'Activities' section shows one activity: 'System Administrator'. The 'Assignment group' is set to 'Software'. The form has buttons for 'Update', 'Close Task', and 'Delete' at the bottom.

Figure: Software Fulfillment Catalog Task - SCTASK0010009

4.7 Email Validation

ServiceNow email records generated during request processing were reviewed to verify request and approval-related communication.

4.8 End-to-End Testing

The complete scenario is tested using a real software installation request.

Employee submits request

↓

REQ/RITM created

↓

License availability checked

↓

Manager approval requested

↓

Approval decision recorded

↓

Catalog Task created

↓

Software Support group

↓

Task processing

Successful completion of this scenario confirms that the major components work together correctly.

4.9 Testing Results

The testing phase confirmed that:

- The Software Installation Request catalog item is available.
- REQ/RITM records are generated.
- Approval processing executes through Flow Designer.
- The approval condition controls the next workflow step.
- A Catalog Task is automatically created after approval.
- The generated task is assigned to the Software support group.
- The Software Request Workflow executes successfully in the target instance.

Phase 5 – Deployment, Documentation & Final Presentation

The fifth and final phase focuses on preparing the **Licensed Software Installation Request** application for deployment, documenting the completed implementation, and presenting the final solution. This phase ensures that the configured ServiceNow components are properly packaged, validated, and ready for demonstration or transfer to another environment.

5.1 Deployment Preparation

Before deployment, the completed configuration is reviewed to ensure that all required components were included.

- Service Catalog Item
- Catalog Variables
- Flow Designer configuration
- Approval configuration
- Catalog Task configuration
- Update Set
- Related request configuration

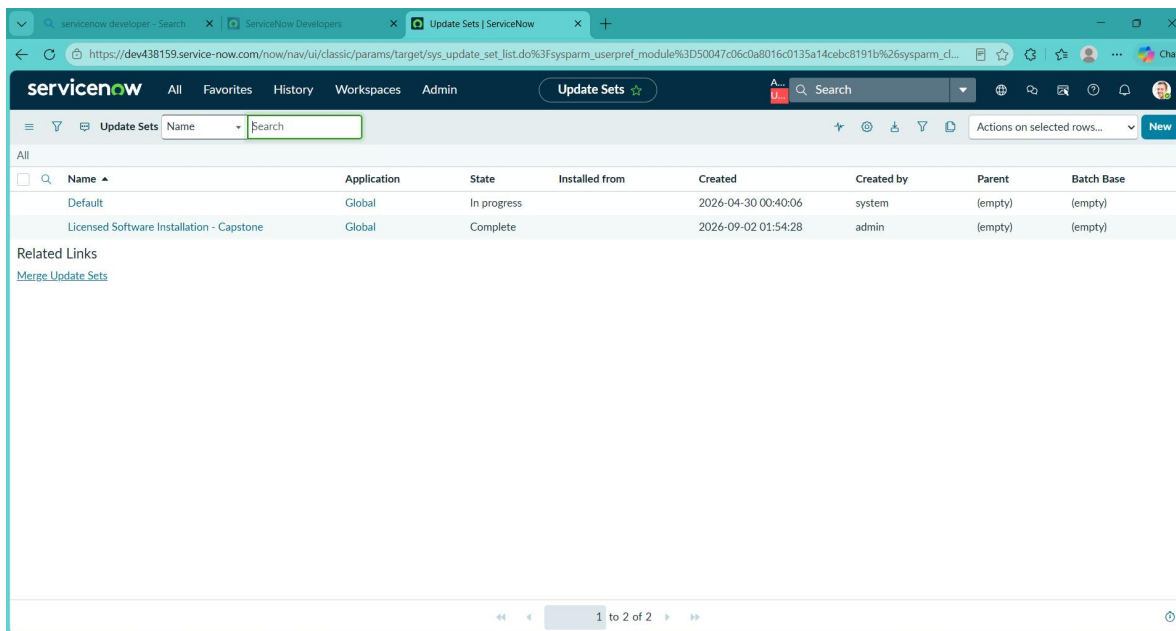
5.2 Update Set Configuration

The project configuration is captured using the ServiceNow Update Set:

Licensed Software Installation - Capstone

The update set provides a controlled method for tracking configuration changes and moving the developed solution between ServiceNow instances.

The project configuration was captured in the **Licensed Software Installation - Capstone** Update Set. The completed Update Set was used to transfer the project configuration from the source instance to the target instance.



The screenshot shows the ServiceNow 'Update Sets' page. The table lists the following data:

Name	Application	State	Installed from	Created	Created by	Parent	Batch Base
Default	Global	In progress		2026-04-30 00:40:06	system	(empty)	(empty)
Licensed Software Installation - Capstone	Global	Complete		2026-09-02 01:54:28	admin	(empty)	(empty)

Below the table, there is a 'Related Links' section with a link to 'Merge Update Sets'.

Figure 5.1: Licensed Software Installation – Capstone – Completed Update Set

5.3 Update Set Deployment

The completed Update Set was exported from the source instance and imported into the target instance. The imported Update Set was previewed, the preview problems were resolved, and the Update Set was successfully committed.

Name	Application	State	Update source	Description	Loaded	Committed	Parent	Remote Batch Base
Licensed Software Installation - Capstone	Global	Committed	(empty)	Update Set for Employee Request Installa...	2026-09-17 11:24:06	2026-09-17 23:56:51	(empty)	(empty)

Related Links

[Import Update Set from XML](#)

Figure 5.2: Licensed Software Installation – Capstone – Committed Update Set

5.4 Post-Deployment Validation

After deployment, the transferred configuration was validated in the target instance. The Software Installation Request catalog item and Software Request Workflow were verified, The deployment process includes:

Development → Testing → Update Set Completion → Export/Retrieve → Target Instance → Validation

The retrieved update set is reviewed before committing the changes to the target environment.

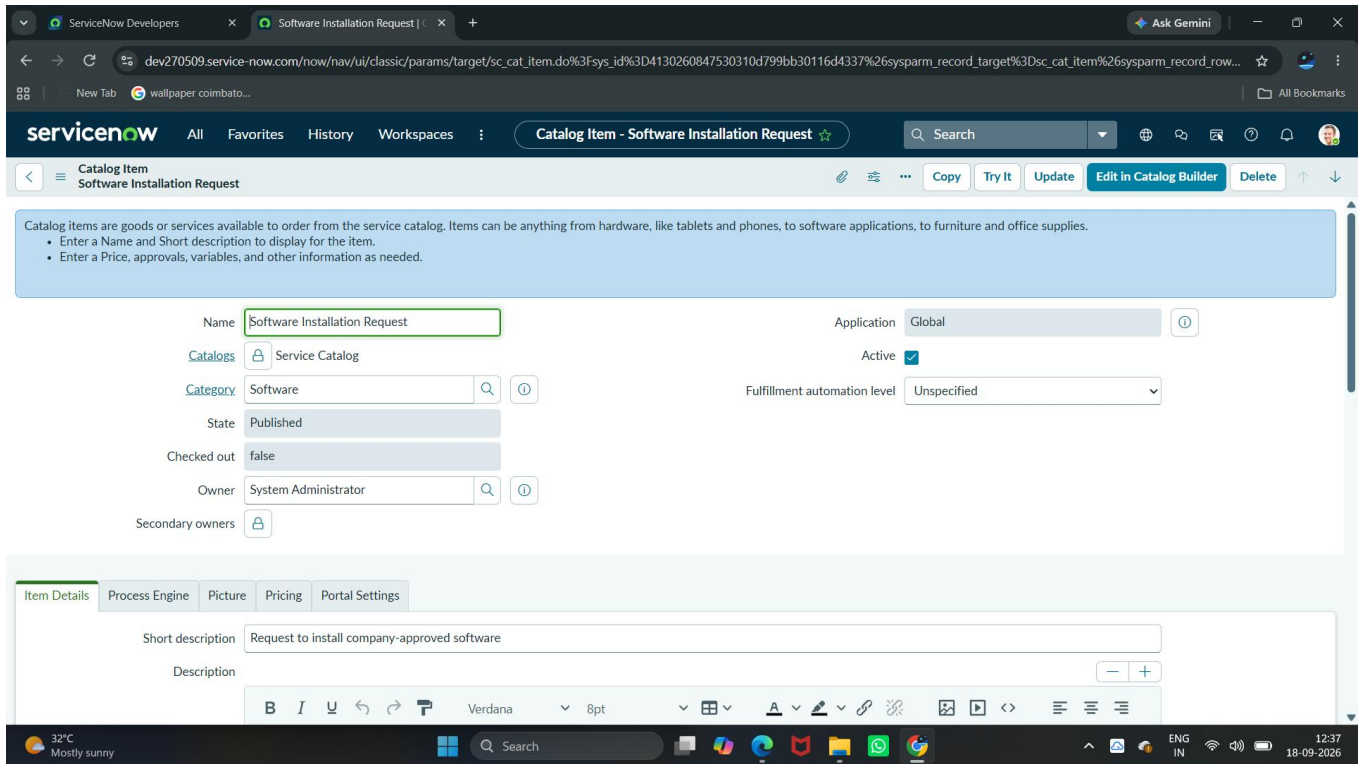


Figure 5.3: Software Installation Request - Target Instance Validation

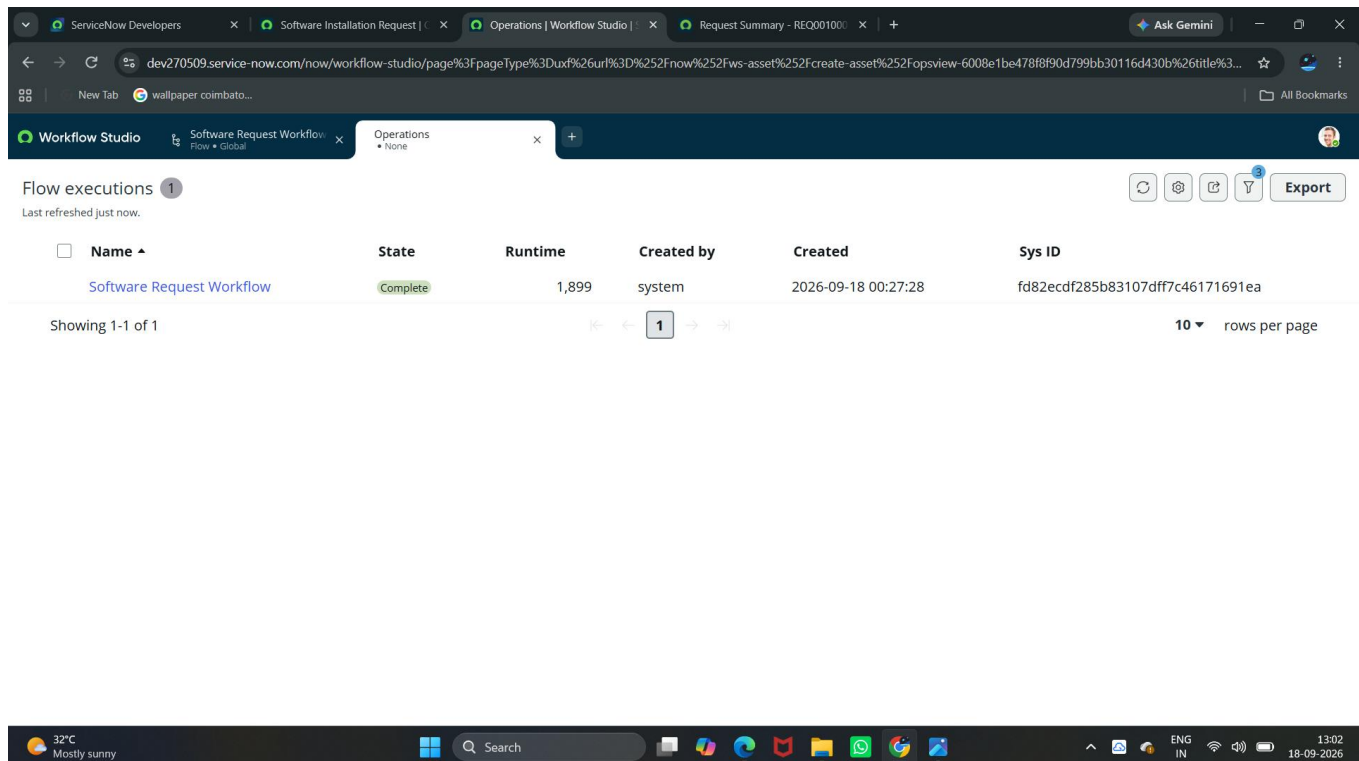


Figure 5.4: Software Request Workflow - Successful Target-Instance Execution

5.5 Documentation

The project documentation records the complete implementation process, including:

Phase 1 – Requirement Analysis & Planning

Requirement identification, stakeholders, functional requirements, and project planning.

Phase 2 – Service Catalog & Form Configuration

Catalog item, variables, form configuration, validation, and REQ/RITM generation.

Phase 3 – Workflow Automation & Approval

Form layout, mandatory fields, and user experience improvements. Workflow, license validation, approval process, catalog task creation, notifications.

Phase 4 – Testing & Validation

Functional testing, workflow testing, approval testing, notification testing, SLA testing, security testing, and end-to-end validation.

Phase 5 – Deployment, Documentation & Final Presentation

Update Set, deployment, dashboard, final validation, documentation, and project presentation.

5.6 Final Demonstration Flow

The final presentation demonstrates the complete employee software request lifecycle:

Employee



Service Catalog



Licensed Software Installation Request



REQ / RITM Creation



Approval



Catalog Task



Software Support Group



Task Processing



Status Tracking

5.6 Final Project Deliverables

The completed project provides:

- Software Installation Request Catalog Item
- Configured Catalog Variables
- Software Request Workflow
- Approval Configuration
- Automated Catalog Task Creation
- Software Support Group Assignment
- Request and Task Tracking
- Update Set Deployment
- Target-Instance Validation
- Project Documentation and Demonstration

5.8 Phase 5 Outcome

At the end of Phase 5, the Software Installation Request solution was documented, tested, packaged using the **Licensed Software Installation - Capstone** Update Set, transferred to a target ServiceNow instance, committed successfully, and validated through a successful flow execution.

Final Workflow – Image Evidence Set

The following screenshots provide visual evidence for the complete process, from catalog request and workflow automation through approval, fulfillment, SLA monitoring, dashboard reporting, and deployment.

The screenshot shows the 'Catalog Item - Software Installation Request' configuration page in ServiceNow. The page includes a header with the ServiceNow logo and navigation tabs. Below the header, there is a section for 'Catalog Item' with various fields and tabs.

Item Details:

- Name:** Software Installation Request
- Application:** Global
- Active:** ☒
- Fulfillment automation level:** Unspecified
- Category:** Software
- State:** Published
- Checked out:** false
- Owner:** System Administrator
- Secondary owners:** (empty)

Short description: Request to install company-approved software

Description: (Rich text editor with formatting options)

Figure 1: Licensed Software Installation Request - Catalog Item Configuration

The screenshot shows the 'Variables' tab for the 'Software Installation Request' catalog item. It displays a table of configured variables with columns for Type, Question, and Order.

Type	Question	Order
Single Line Text	What software do you need?	100
Single Line Text	Specify version (if required)	200
Multi Line Text	Why do you need this software?	300
Select Box	Select urgency level.	400

The table is paginated, showing 1 to 4 of 4 items.

Figure 2: Licensed Software Installation Request - Configured Catalog Variables

The screenshot shows the ServiceNow interface for a 'Software Installation Request'. The breadcrumb trail is 'Home > Service Catalog > Software > Software Installation Request'. The form title is 'Software Installation Request' with the subtitle 'Request to install company-approved software'. A description states: 'Allows employees to request software installations. Fulfills via Software Support Team.' The form contains several input fields: 'What software do you need?' (empty), 'Specify version (if required)' (empty), 'Why do you need this software?' (empty), and 'Select urgency level.' (set to 'Critical'). There is an 'Add attachments' section at the bottom. On the right side, a sidebar shows 'Quantity: 1', 'Delivery Time: 0 Days', and buttons for 'Add to Cart', 'Save as Draft', and 'Order Now'. The top navigation bar includes links for Knowledge, Catalog, Requests, Approvals, System Status, Cart, Tours, and a user profile for 'System Administrator'.

Figure 3: Licensed Software Installation Request - Service Portal Request Form

This screenshot shows the same 'Software Installation Request' form, but with the following fields filled out: 'What software do you need?' contains 'Microsoft Office 365', 'Specify version (if required)' contains '2021', and 'Why do you need this software?' contains 'Required for daily work'. The 'Select urgency level.' dropdown is now set to 'High'. The right sidebar shows 'Quantity: 1' and 'Delivery Time: 2 Days'. The 'Add to Cart', 'Save as Draft', and 'Order Now' buttons are still present. The top navigation bar is identical to the previous screenshot, showing the same user profile and system links.

Figure 4: Licensed Software Installation Request - filled Service Portal Request Form

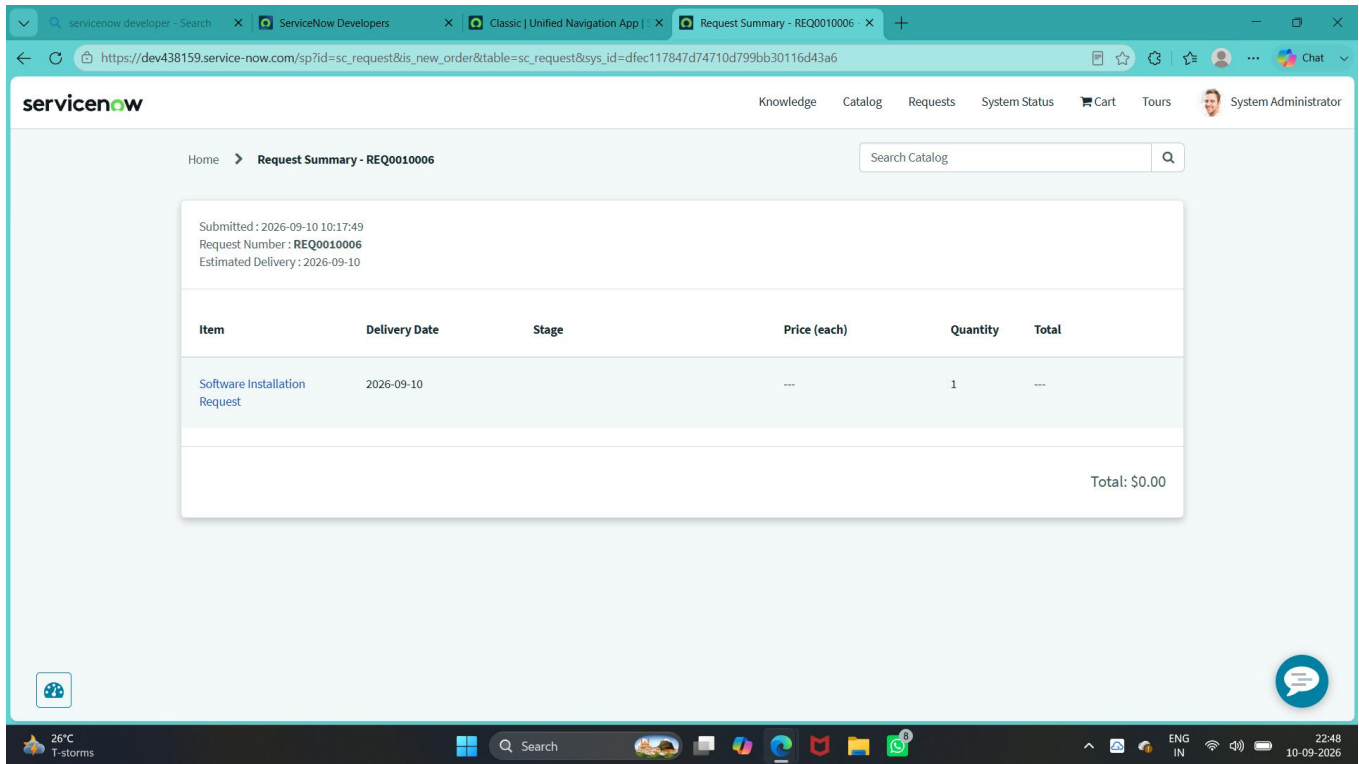


Figure 5: Licensed Software Installation Request - Generated REQ/RITM Record

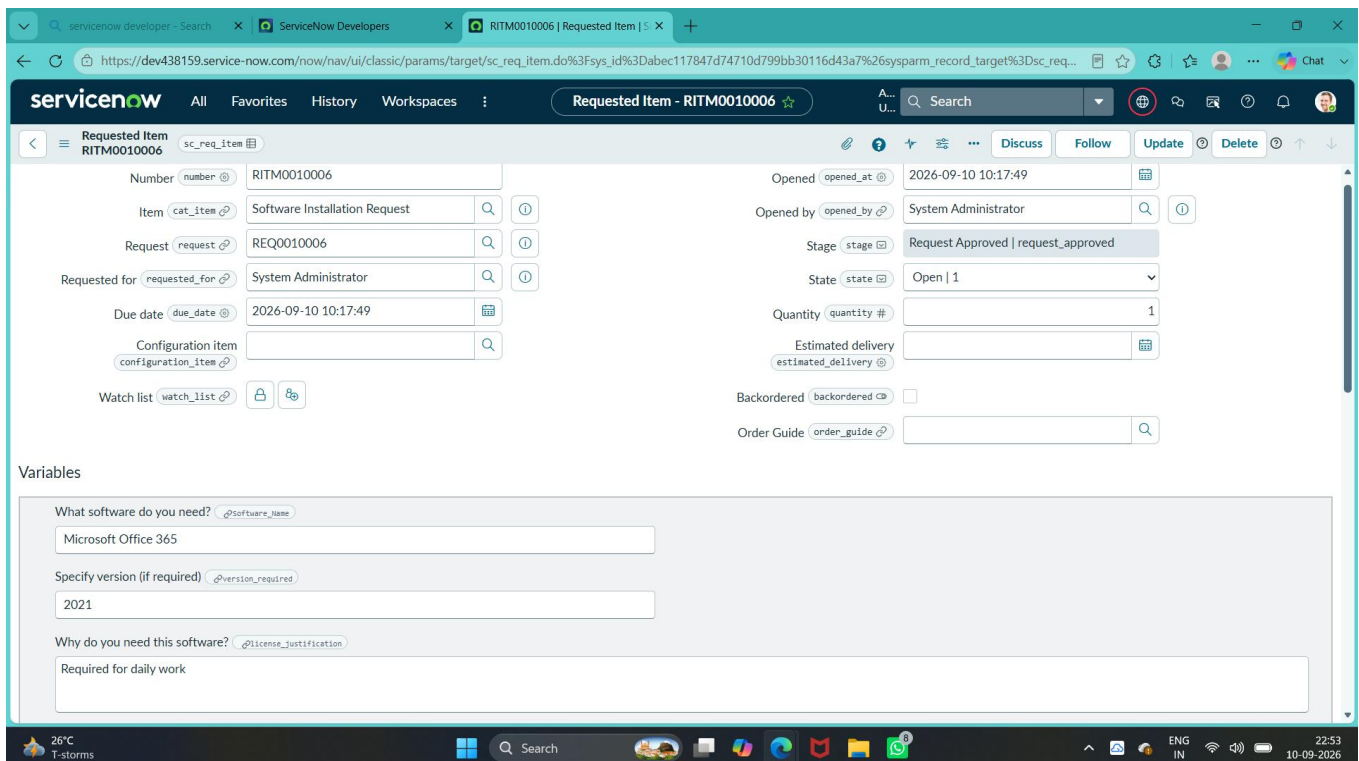


Figure 6: Licensed Software Installation Request - Generated REQ/RITM Record

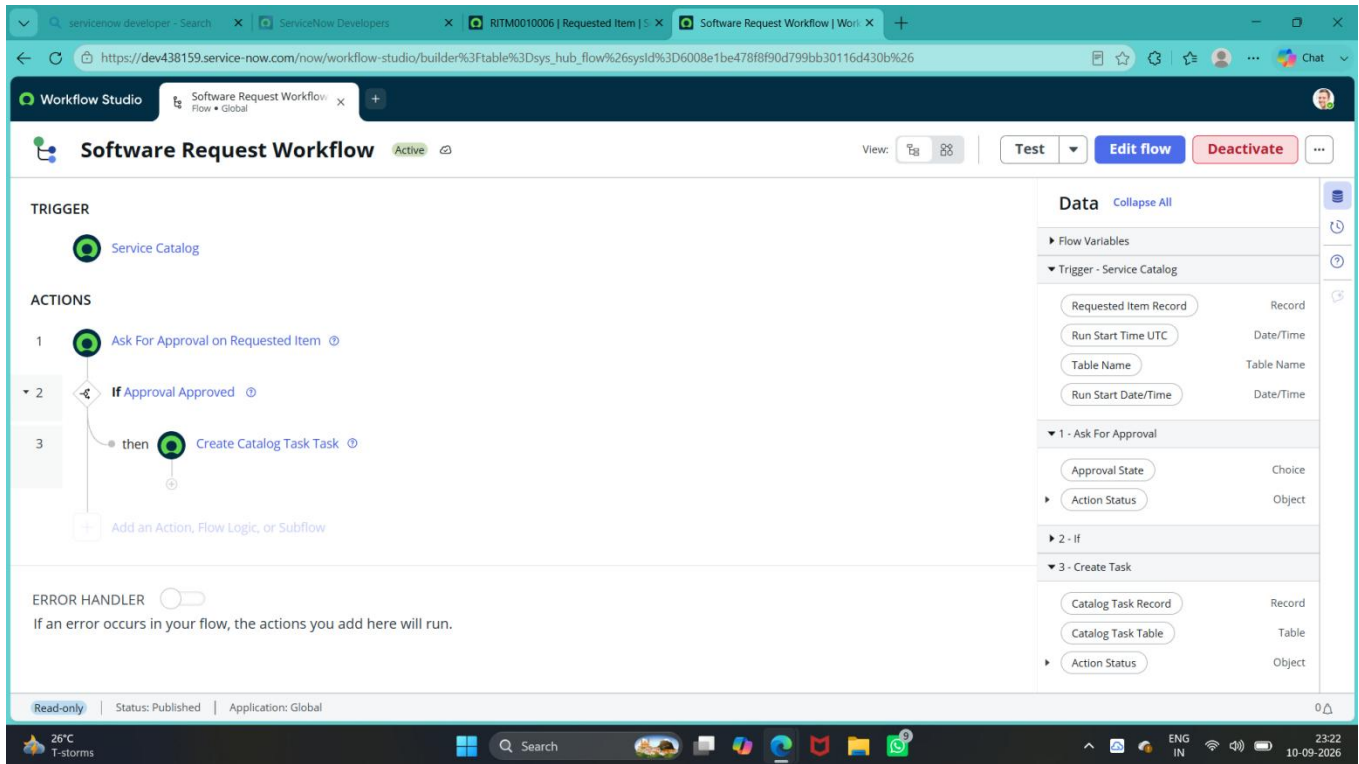


Figure 7: Software Request Workflow - Workflow Automation Configuration

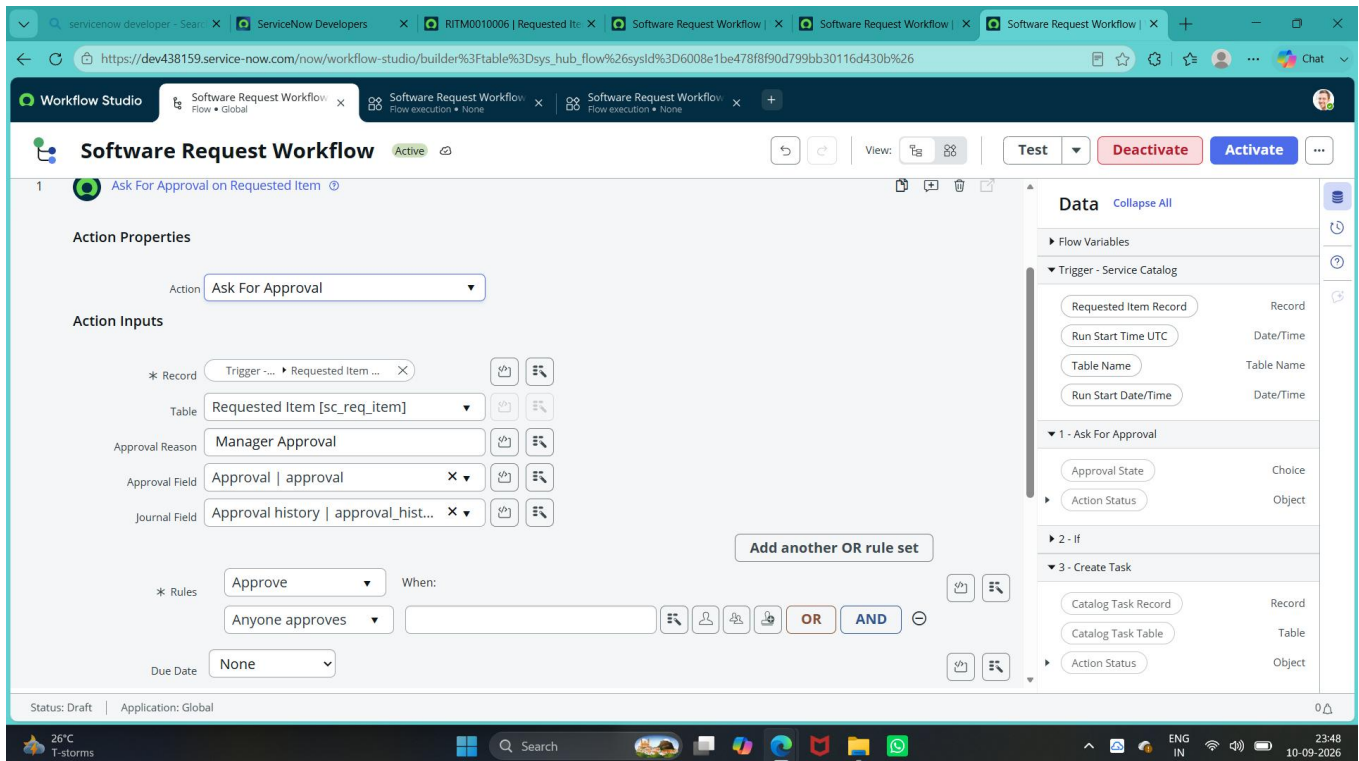


Figure 8: Software Request Workflow - Approval Configuration

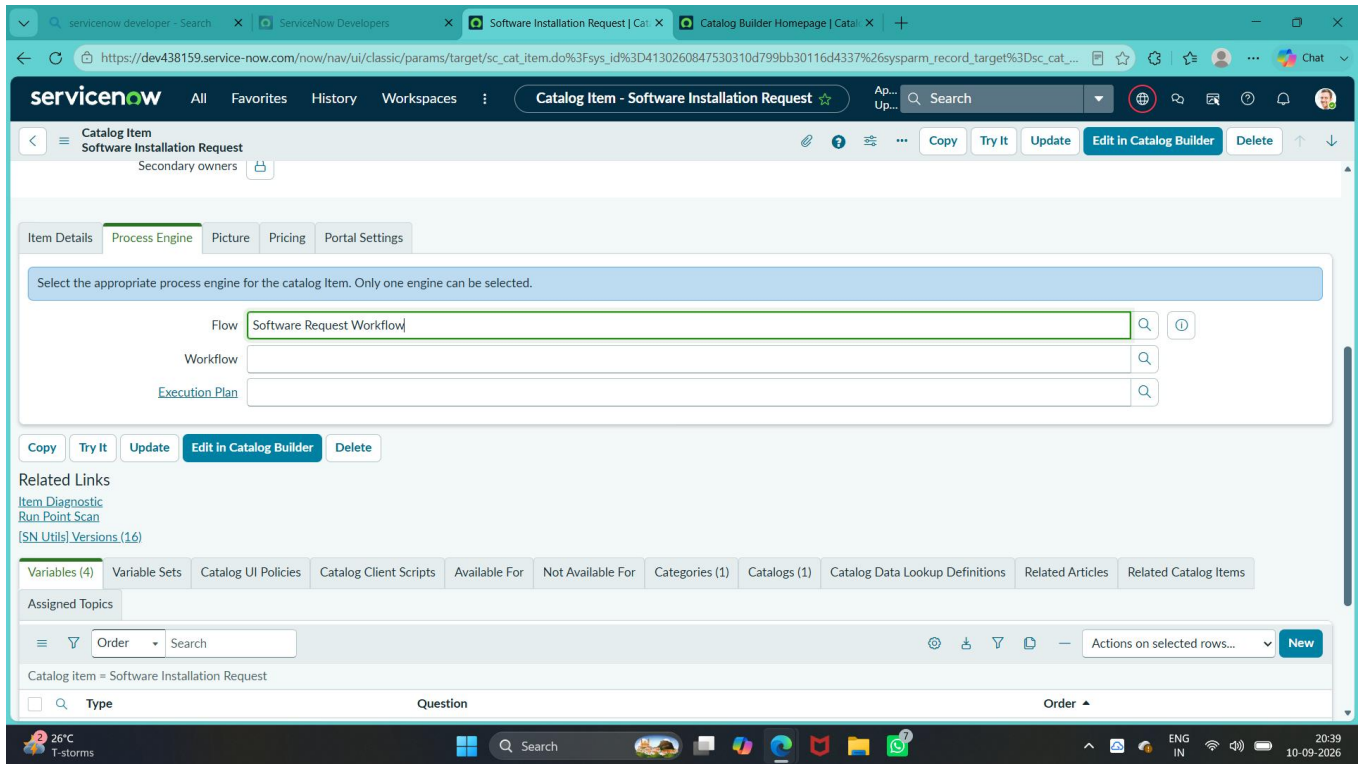


Figure 9: Software Installation Request - Process Engine Configuration

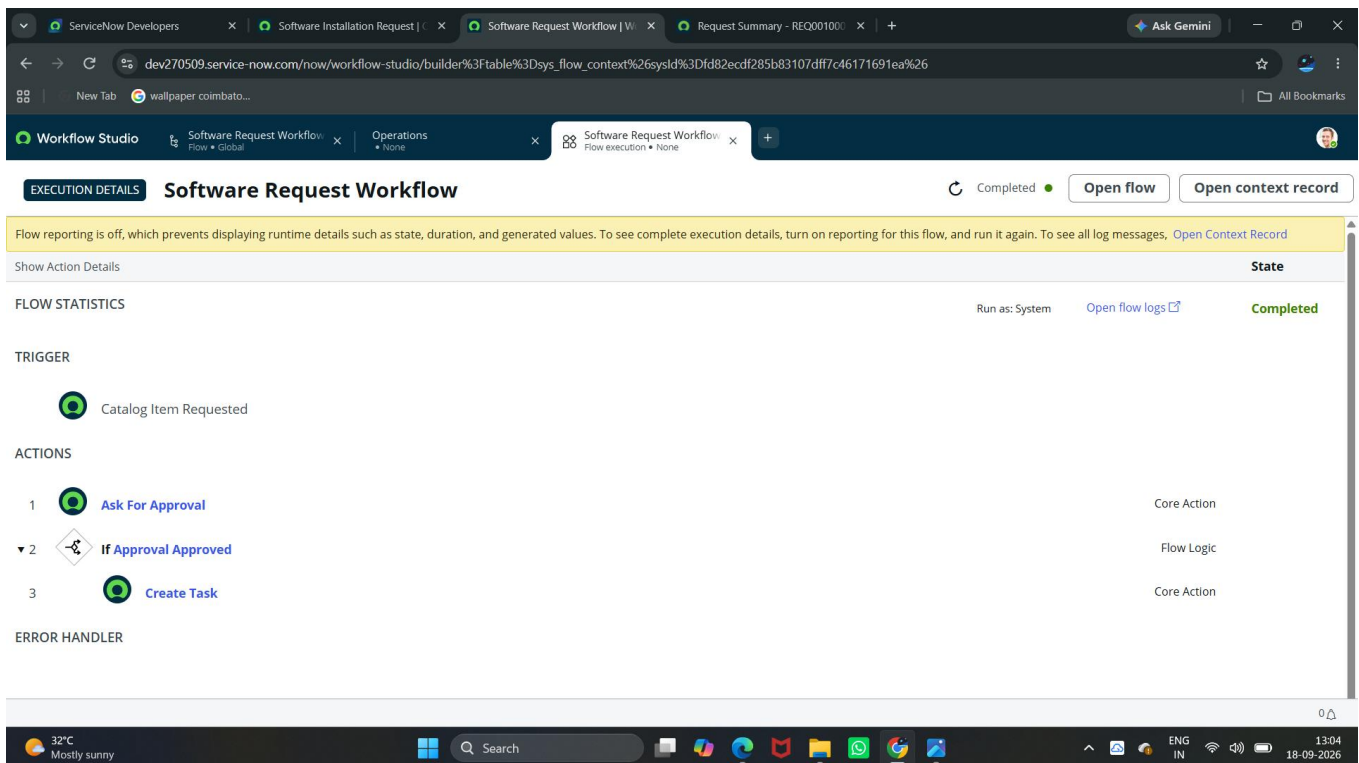


Figure 10: Software Request Workflow - Successful Flow Execution

Number: SCTASK0010009

Assigned to: [Search]

Configuration item: [Search]

Active: ☒

Approval: Not Yet Requested

Priority: -- None --

State: Open

Request item: [Search]

Requested for: [Search]

Short description: Software Fulfillment Task

Description: [Text Area]

Work notes: [Text Area]

Comments (Customer visible): ☐ **Post**

Activities: 1

- System Administrator
 - Impact: 3 - Low
 - Opened by: System Administrator
 - State: Open

Field changes • 2026-09-10 11:24:21

Assignment group: Software

Buttons: Update, Close Task, Delete

Figure 11: Software Fulfillment Task - Catalog Task Configuration

serviceNow All Favorites History Workspaces Admin Update Sets Search

Update Sets Name [Search]

Actions on selected rows... New

Name	Application	State	Installed from	Created	Created by	Parent	Batch Base
Default	Global	In progress		2026-04-30 00:40:06	system	(empty)	(empty)
Licensed Software Installation - Capstone	Global	Complete		2026-09-02 01:54:28	admin	(empty)	(empty)

Related Links

[Merge Update Sets](#)

1 to 2 of 2

Figure 12: Licensed Software Installation - Update Set Completed

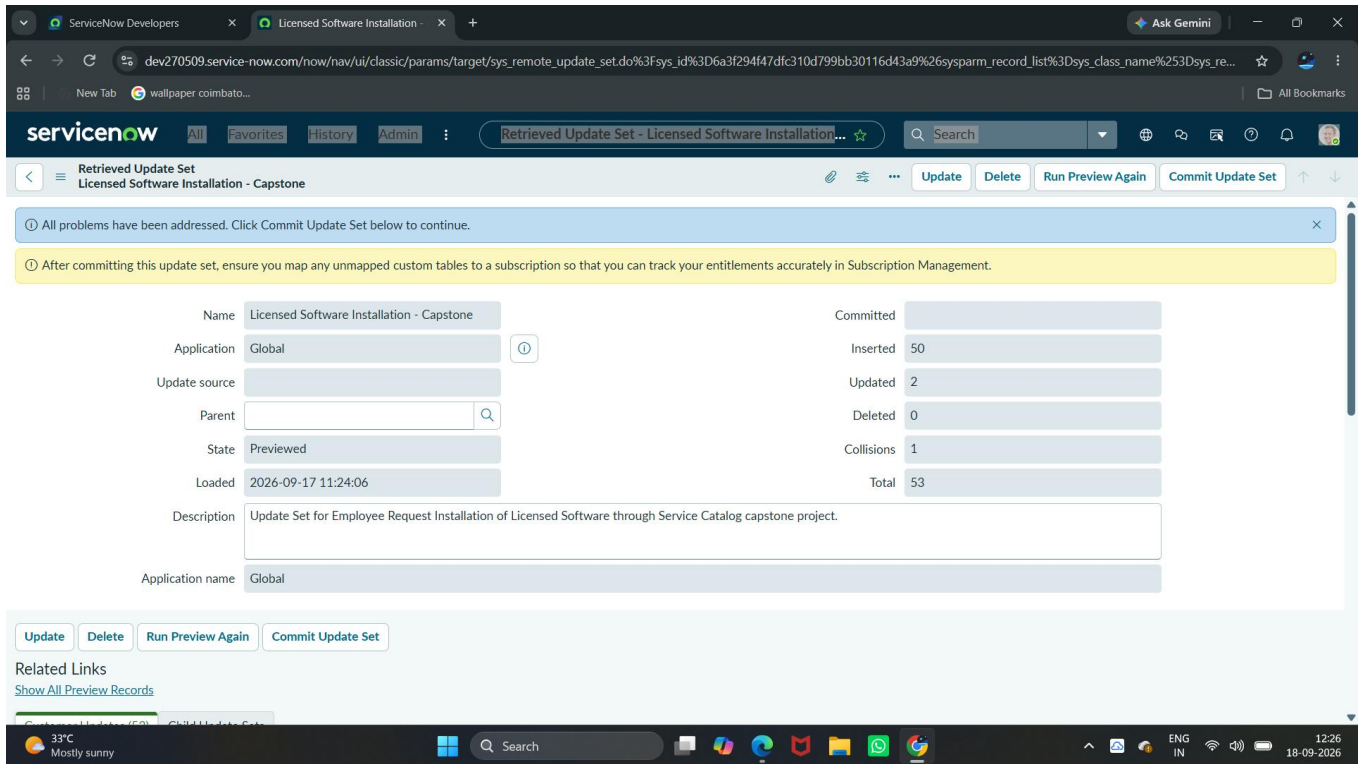


Figure 13: Licensed Software Installation - Update Set Preview Resolved

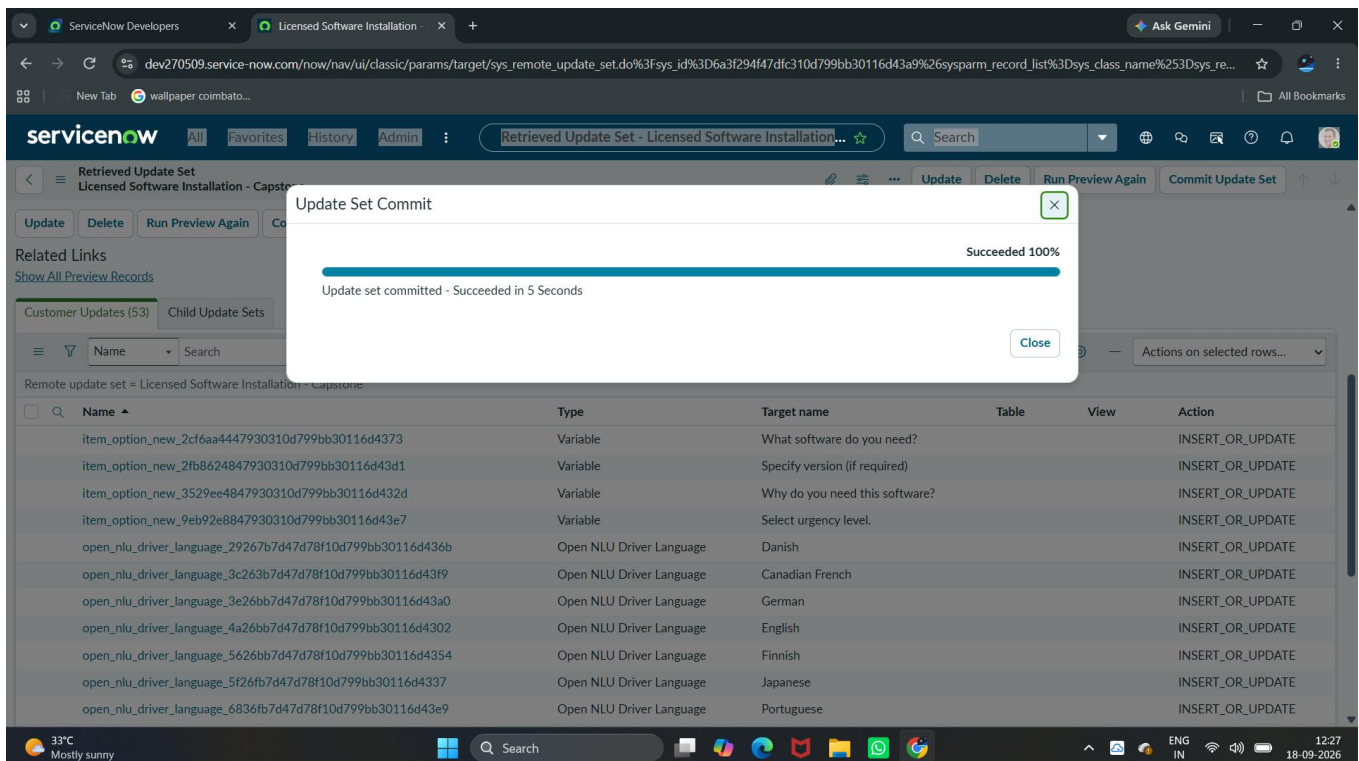


Figure 14: Licensed Software Installation - Update Set Committed

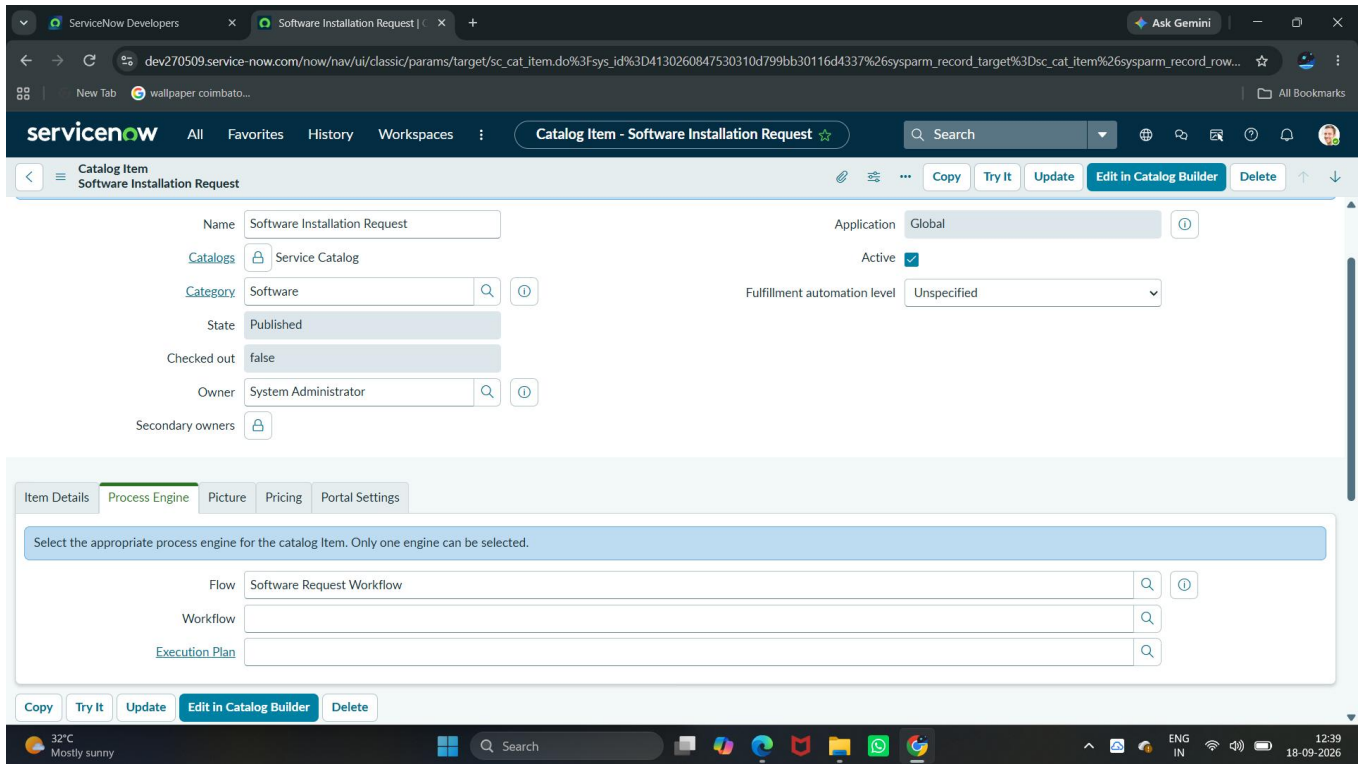


Figure 15: Target Instance - Software Installation Request Catalog Item

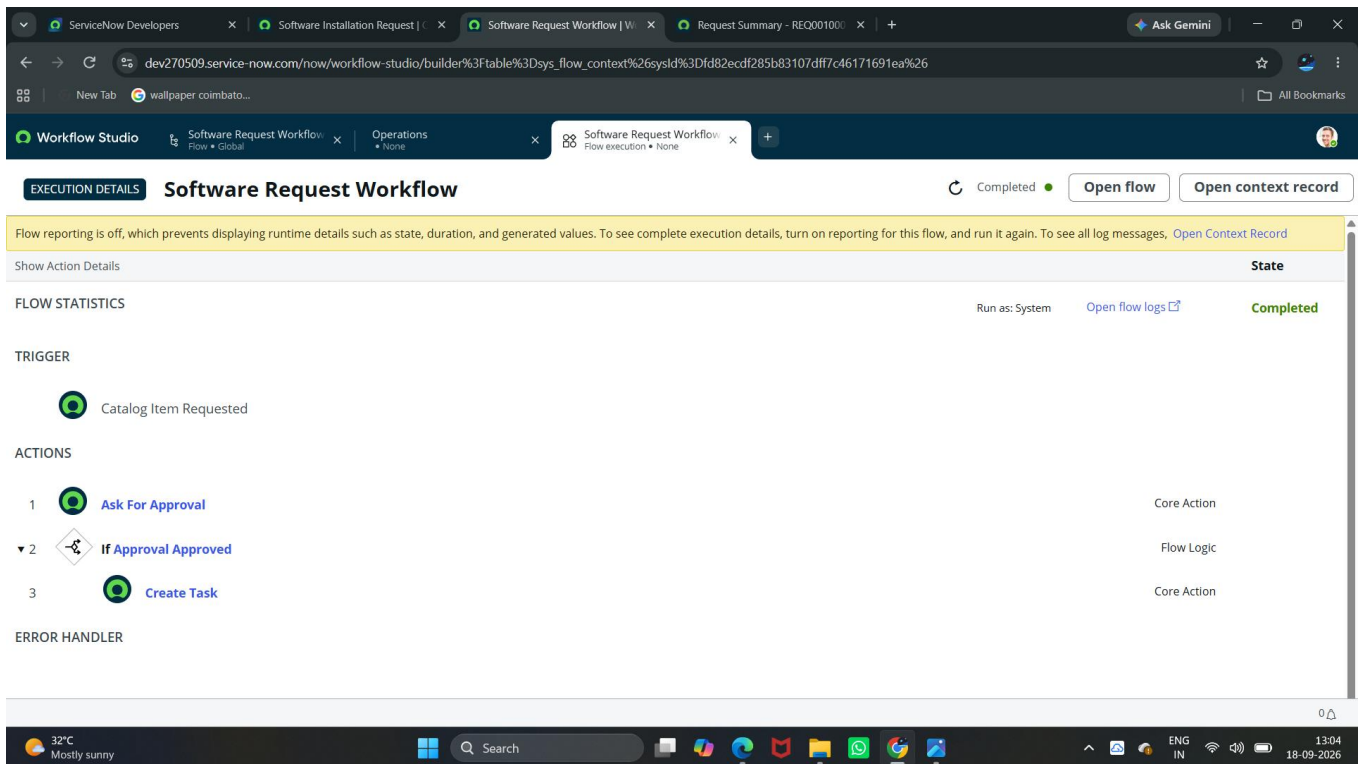


Figure 16: Target Instance - Software Request Workflow Execution

Project Conclusion

The **Automated Licensed Software Installation Request Management** project provides a structured ServiceNow-based process for managing employee software installation requests. Employees can submit a Software Installation Request through the Service Catalog by providing software name, required version, license justification, and urgency.

The request is processed through Flow Designer for approval, and after approval, a Catalog Task is automatically created for the Software support group. The solution provides traceability across REQ, RITM, and SCTASK records and uses Update Sets to transfer the completed configuration between ServiceNow instances.

Overall, the project demonstrates how ServiceNow can automate software request approval and fulfillment while improving request tracking and reducing manual coordination.

Final Project Flow

Employee Request → Service Catalog → REQ/RITM → Approval → Catalog Task → Software Support Group → Task Processing → Status Tracking